Marwadi University
Faculty of Computer Applications

BCA 4E Python Mid-1 Exam

Enrollment No. : 92400527276

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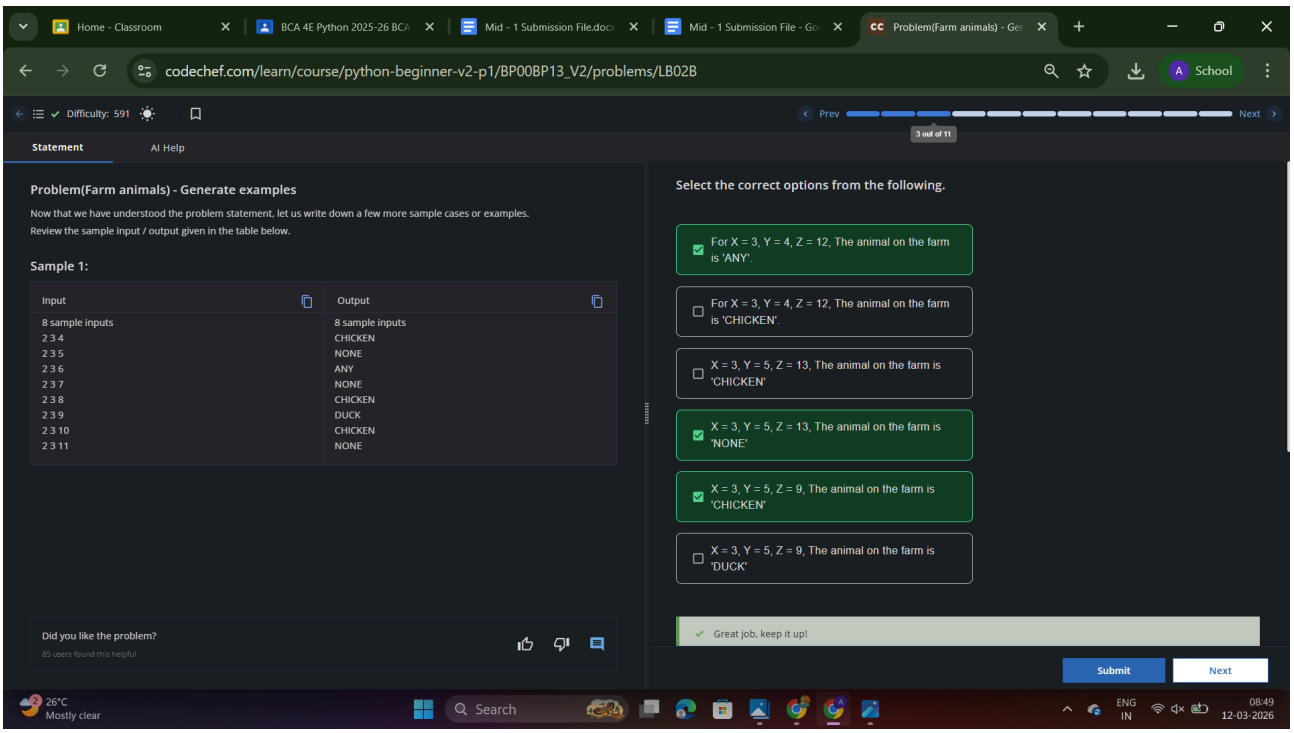
Class :4E

Date : 12-03-2026



Answer – 1

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codechef.com/learn/course/python-beginner-v2-p1/BP00BP13_V2/problems/LB02D?tab=statement

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Problem(Farm animals) - Solve sub-components

How do we find the following?

- Z is divisible only by X
- Z is divisible only by Y
- Z is divisible by both X and Y
- Z is divisible by neither X nor Y

Code it out in the IDE and click on **Submit** to continue.

Did you like the problem?

25 users found this helpful

Python3

```python
X = 3
Y = 4
Z = 12
if (Z % X == 0) and (Z % Y == 0):
    print('Z is divisible by both X and Y')
elif (Z % X == 0):
    print('Z is divisible only by X')
elif (Z % Y == 0):
    print('Z is divisible only by Y')
else:
    print('Z is divisible by neither X nor Y')

X = 3
Y = 5
Z = 13
if (Z % X == 0) and (Z % Y == 0):
    print('Z is divisible by both X and Y')
elif (Z % X == 0):
    print('Z is divisible only by X')
elif (Z % Y == 0):
    print('Z is divisible only by Y')
else:
    print('Z is divisible by neither X nor Y')
```

Your Output

```
Z is divisible by both X and Y
Z is divisible by neither X nor Y
```

Excellent work!

Visualize Code | Submit | Next

26°C Mostly clear | Search | ENG IN | 08:51 12-03-2026

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Problem(Farm animals) - Solve the complete problem

In Chefland, each chicken has X legs and each duck has Y legs.
Chef's farm can have **exactly one type** of bird.
Given that the birds on the farm have a total of Z legs:

- Print `CHICKEN`, if the farm can have only chickens but not ducks.
- Print `DUCK`, if the farm can have only ducks but not chickens.
- Print `ANY`, if the farm can have either chickens or ducks.
- Print `NONE`, if the farm can have neither chickens nor ducks.

Input format

- The first line will contain T, the number of test cases. Then the test cases follow.
- Each test case consists of a single line of input, containing three space-separated integers X, Y, Z.

Sample 1:

Input	Output
3 2 3 5 2 2 2 3 4 6	NONE ANY CHICKEN

Explanation:

Test case 1: There are 5 legs on the farm. Each chicken has 2 legs and each duck has 3 legs. Thus, the farm can have neither chickens nor ducks.

Test case 2: There are 2 legs on the farm. Each chicken has 2 legs and each duck has 2 legs. Thus, the farm can have any bird among chicken and duck.

Python3

```python
# Solution as follows

t = int(input())
for i in range(t):
    X, Y, Z = map(int, input().split())
    if (Z%X == 0) and (Z%Y == 0):
        print('any')
    elif (Z%X == 0):
        print('chicken')
    elif (Z%Y == 0):
        print('duck')
    else:
        print('none')
```

Test against Custom Input

```
3
2 3 5
2 2 2
3 4 6
```

Visualize Code | Run | Submit | Next

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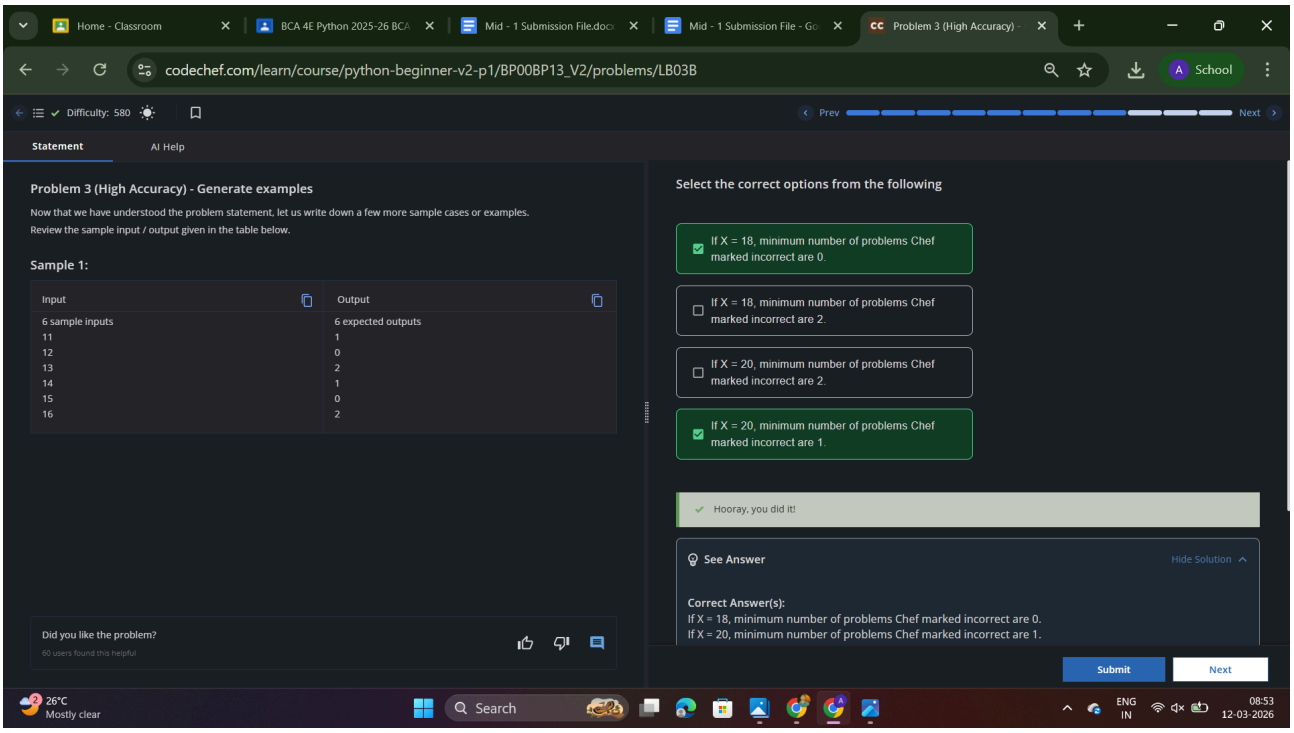
Answer – 2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Problem 3 (High Accuracy) -
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Problem 3 (High Accuracy) - Generate examples
Now that we have understood the problem statement, let us write down a few more sample cases or examples.
Review the sample input / output given in the table below.
Sample 1:
Input
6 sample inputs
11
12
13
14
15
16
Output
6 expected outputs
1
0
2
1
0
2
Did you like the problem?
60 users found this helpful
Select the correct options from the following
If X = 18, minimum number of problems Chef marked incorrect are 0.
If X = 18, minimum number of problems Chef marked incorrect are 2.
If X = 20, minimum number of problems Chef marked incorrect are 2.
If X = 20, minimum number of problems Chef marked incorrect are 1.
Hooray, you did it!
See Answer
Hide Solution
Correct Answer(s):
If X = 18, minimum number of problems Chef marked incorrect are 0.
If X = 20, minimum number of problems Chef marked incorrect are 1.
Submit
Next
26°C
Mostly clear
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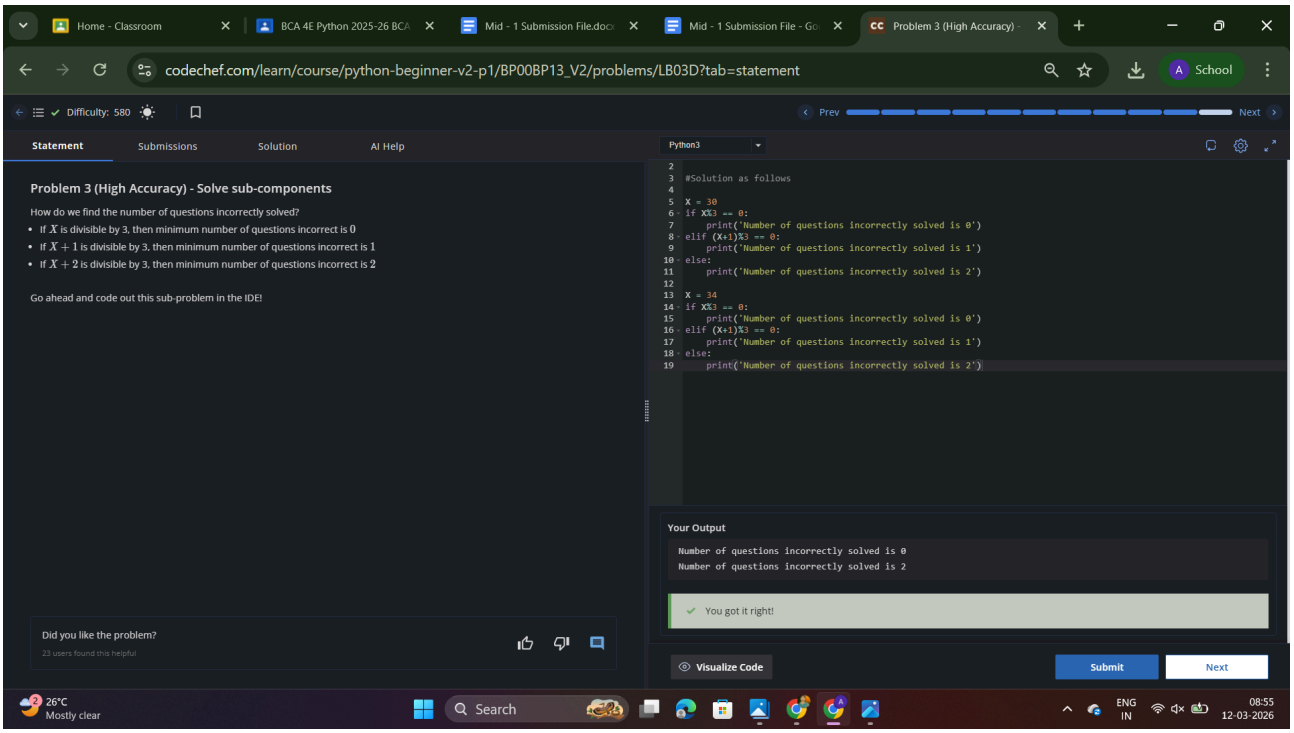
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School
Difficulty: 580
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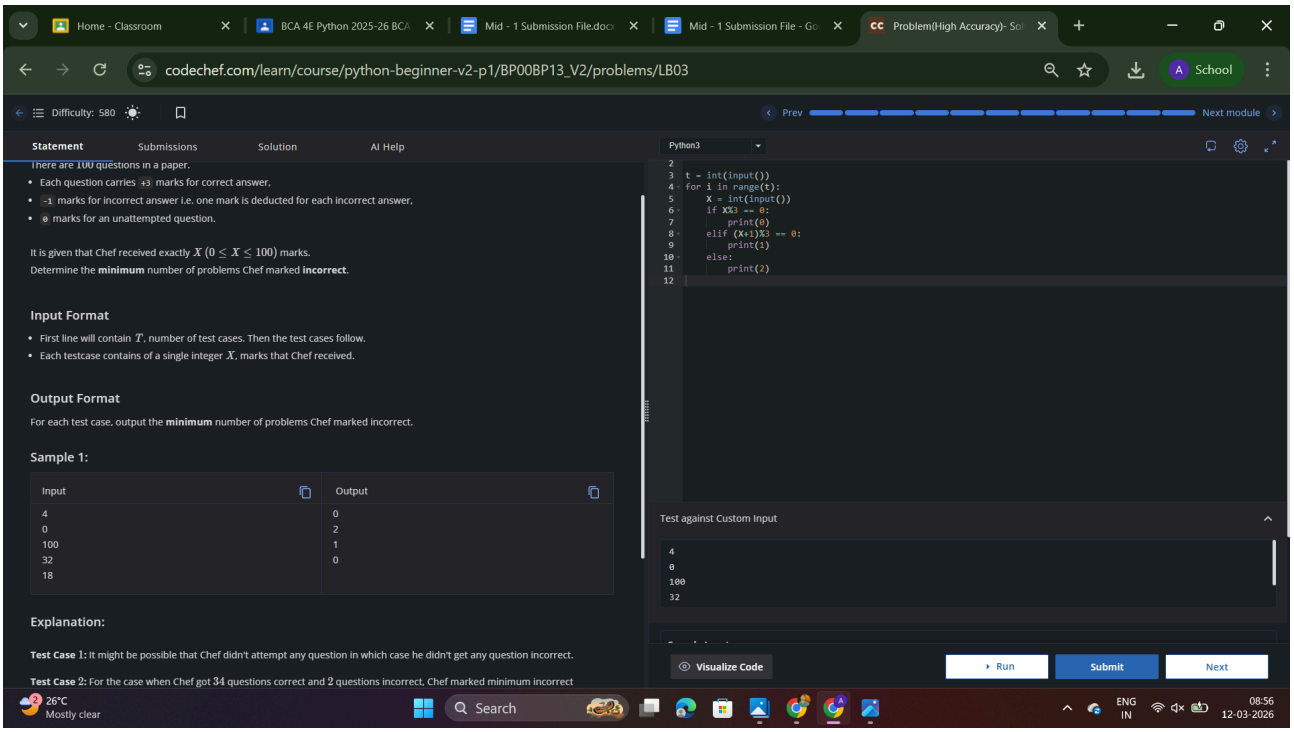
Problem 3 (High Accuracy) - Solve sub-components
How do we find the number of questions incorrectly solved?
If X is divisible by 3, then minimum number of questions incorrect is 0
If $X+1$ is divisible by 3, then minimum number of questions incorrect is 1
If $X+2$ is divisible by 3, then minimum number of questions incorrect is 2
Go ahead and code out this sub-problem in the IDE!
Did you like the problem?
23 users found this helpful
Python3
#Solution as follows
X = 30
if X%3 == 0:
print('Number of questions incorrectly solved is 0')
elif (X+1)%3 == 0:
print('Number of questions incorrectly solved is 1')
else:
print('Number of questions incorrectly solved is 2')
X = 34
if X%3 == 0:
print('Number of questions incorrectly solved is 0')
elif (X+1)%3 == 0:
print('Number of questions incorrectly solved is 1')
else:
print('Number of questions incorrectly solved is 2')
Your Output
Number of questions incorrectly solved is 0
Number of questions incorrectly solved is 2
You got it right!
Visualize Code
Submit
Next
26°C
Mostly clear
Search
ENG
IN
08:55
12-03-2026

Answer – 3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Problem(High Accuracy)- Sol
codechef.com/learn/course/python-beginner-v2-p1/BP00BP13_V2/problems/LB03
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Difficulty: 580
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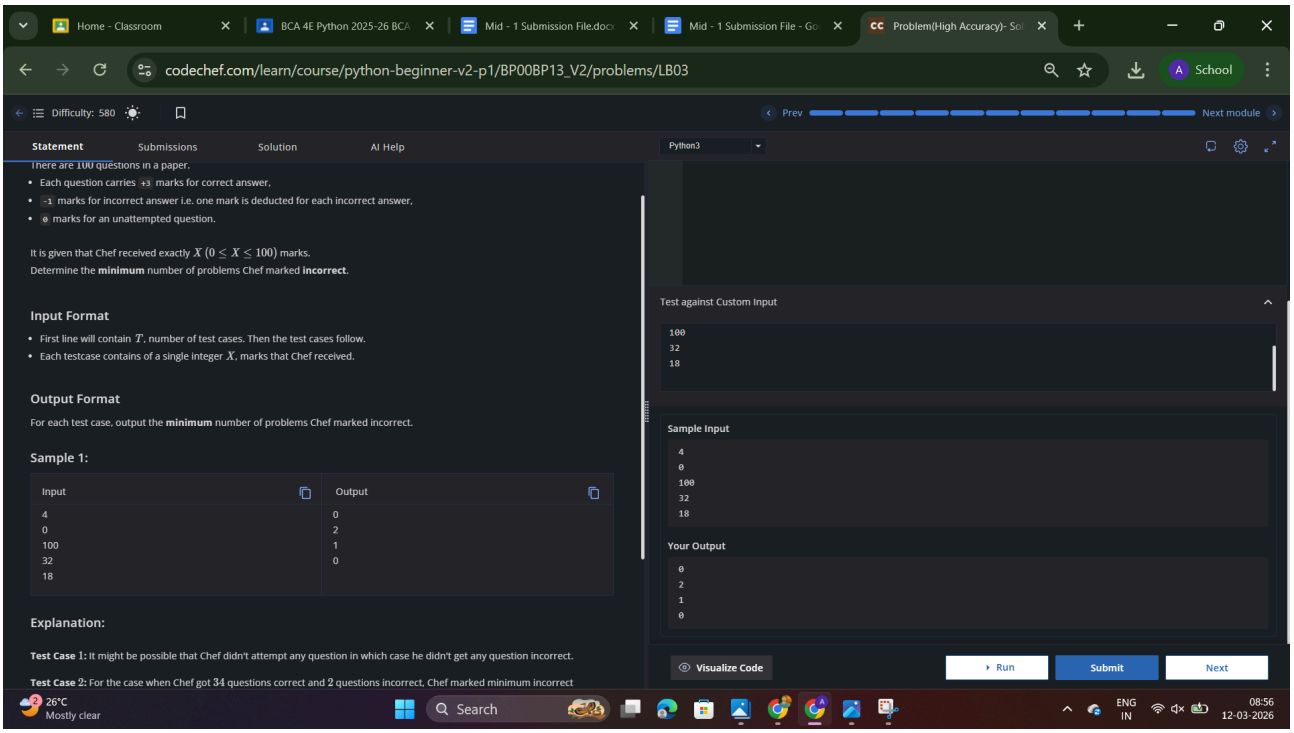
There are 100 questions in a paper.
Each question carries +3 marks for correct answer,
-1 marks for incorrect answer i.e. one mark is deducted for each incorrect answer,
0 marks for an unattempted question.
It is given that Chef received exactly X $(0 \le X \le 100)$ marks.
Determine the **minimum** number of problems Chef marked **incorrect**.
Input Format
First line will contain T, number of test cases. Then the test cases follow.
Each testcase contains of a single integer X, marks that Chef received.
Output Format
For each test case, output the **minimum** number of problems Chef marked incorrect.
Sample 1:
Input
4
0
100
32
18
Output
0
2
1
0
Explanation:
Test Case 1: It might be possible that Chef didn't attempt any question in which case he didn't get any question incorrect.
Test Case 2: For the case when Chef got 34 questions correct and 2 questions incorrect, Chef marked minimum incorrect
Python3
t = int(input())
for i in range(t):
X = int(input())
if X%3 == 0:
print(0)
elif (X+1)%3 == 0:
print(1)
else:
print(2)
Test against Custom Input
4
0
100
32
Visualize Code
Run
Submit
Next
26°C
Mostly clear
Search
ENG
IN
08:56
12-03-2026



Difficulty: 580

Statement | Submissions | Solution | AI Help

There are 100 questions in a paper.

- Each question carries `+3` marks for correct answer,
- `-1` marks for incorrect answer i.e. one mark is deducted for each incorrect answer,
- `0` marks for an unattempted question.

It is given that Chef received exactly X $(0 \leq X \leq 100)$ marks.
Determine the **minimum** number of problems Chef marked **incorrect**.

Input Format

- First line will contain T, number of test cases. Then the test cases follow.
- Each testcase contains of a single integer X, marks that Chef received.

Output Format

For each test case, output the **minimum** number of problems Chef marked incorrect.

Sample 1:

Input	Output
4 0 100 32 18	0 2 1 0

Explanation:

Test Case 1: It might be possible that Chef didn't attempt any question in which case he didn't get any question incorrect.

Test Case 2: For the case when Chef got 34 questions correct and 2 questions incorrect, Chef marked minimum incorrect

Python3

Test against Custom Input

```
100
32
18
```

Sample Input

```
4
0
100
32
18
```

Your Output

```
0
2
1
0
```

Visualize Code | Run | Submit | Next

Answer – 4

Answer - 5